Q1.

A vector is given by $\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix}$

Which vector is **not** perpendicular to **a**?

Circle your answer.

$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} 5 \\ -1 \\ 3 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

(Total 1 mark)

Q2.

The points A, B and C have coordinates (3, −2, 4), (1, −5, 6) and (−4, 5, −1) respectively.

The line l passes through A and has equation $\mathbf{r} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 7 \\ -7 \\ 5 \end{bmatrix}$.

(a) Show that the point C lies on the line l.

(2)

(b) Find a vector equation of the line that passes through points A and B.

(3)

(c) The point D lies on the line through A and B such that the angle CDA is a right angle.

Find the coordinates of D.

(5)

(d) The point E lies on the line through A and B such that the area of triangle ACE is three times the area of triangle ACD.

Find the coordinates of the two possible positions of E.

(4)

(Total 14 marks)

Q3.

A line and a plane have equations

$$\frac{x-3}{p} = \frac{y-q}{3} = \frac{z-1}{-1}$$

and $\mathbf{r} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 10$

respectively, where p and q are constants.

(a) Show that the line is **not** perpendicular to the plane.

(1)

(b) In the case where the line lies in the plane, find the values of p and q.

(4)

(c) In the case where the angle, θ, between the line and the plane satisfies $\sin\theta = \frac{1}{\sqrt{6}}$, and the line intersects the plane at $z = 2$:

(i) find the value of p;

(5)

(ii) find the value of q.

(2)

(Total 12 marks)

Q4.

The line l_1 has equation $\mathbf{r} = \begin{bmatrix} 0 \\ -2 \\ q \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$, where q is an integer.

The line l_2 has equation $\mathbf{r} = \begin{bmatrix} 8 \\ 3 \\ 5 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$.

The lines l_1 and l_2 intersect at the point P.

(a) Show that $q = 4$ and find the coordinates of P.

(3)

(b) Show that l_1 and l_2 are perpendicular.

(1)

(c) The point A lies on the line l_1 where $\lambda = 1$.

(i) Find AP^2.

(2)

(ii) The point B lies on the line l_2 so that the right-angled triangle APB is isosceles.

Find the coordinates of the two possible positions of B.

(6)
(Total 12 marks)

Q5.

The lines L_1 and L_2 have equations

$$\mathbf{r} = \begin{bmatrix} 7 \\ -25 \\ 9 \end{bmatrix} + \alpha \begin{bmatrix} 3 \\ -4 \\ 7 \end{bmatrix} \text{ and } \mathbf{r} = \begin{bmatrix} 7 \\ 19 \\ -2 \end{bmatrix} + \beta \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$$

respectively.

(a) Determine a vector, $\mathbf{n}$, which is perpendicular to both lines.

(2)

(b) (i) The point A on L_1 and the point B on L_2 are such that $\overrightarrow{AB} = \lambda\mathbf{n}$ for some constant λ.

Show that

$$3\alpha - 2\beta + 2\lambda = 0$$

$$4\alpha - 2\beta - 5\lambda = -44$$

$$7\alpha - 3\beta + 2\lambda = -11$$

(3)

(ii) Find the position vectors of A and B.

(3)

(iii) Deduce the shortest distance between L_1 and L_2.

(2)
(Total 10 marks)

Q6.

The points A, B and C have coordinates (2, –1, –5), (0, 5, –9) and (9, 2, 3) respectively.

The line l has equation $\mathbf{r} = \begin{bmatrix} 2 \\ -1 \\ -5 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$.

(a) Verify that the point B lies on the line l.

(2)

(b) Find the vector $\overrightarrow{BC}$.

(2)

(c) The point D is such that $\overrightarrow{AD} = 2\overrightarrow{BC}$.

(i) Show that D has coordinates (20, –7, 19).

(2)

(ii) The point P lies on l where $\lambda = p$. The line PD is perpendicular to l. Find the value of p.

(5)
(Total 11 marks)

Q7.

(a) Find the value of p for which the planes with equations

$$\mathbf{r}.\begin{bmatrix}6\\-3\\2\end{bmatrix} = 42 \text{ and } \mathbf{r}.\begin{bmatrix}4p+1\\p-2\\1\end{bmatrix} = -7$$

(i) are perpendicular;

(3)

(ii) are parallel.

(3)

(b) In the case when $p = 4$:

(i) write down a cartesian equation for each plane;

(2)

(ii) find, in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, an equation for l, the line of intersection of the planes.

(6)

(c) Determine a vector equation, in the form $\mathbf{r} = \mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$, for the plane which contains l and which passes through the point (30, 7, 30).

(2)
(Total 16 marks)

Q8.

The points A and B have coordinates (2, 1, –1) and (3, 1, –2) respectively.
The angle OBA is θ, where O is the origin.

(a) (i) Find the vector $\overrightarrow{AB}$.

(2)

(ii) Show that $\cos\theta = \frac{5}{2\sqrt{7}}$.

(4)

(b) The point C is such that $\overrightarrow{OC} = 2\overrightarrow{OB}$. The line l is parallel to $\overrightarrow{AB}$ and passes through the point C. Find a vector equation of l.

(2)

(c) The point D lies on l such that angle $ODC = 90°$. Find the coordinates of D.

(4)

(Total 12 marks)

Q9.

The plane Π has equation $\mathbf{r} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} + \mu \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix}$.

(a) Find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = d$.

(4)

(b) Show that the line with equation $\mathbf{r} = \begin{bmatrix} 7 \\ 1 \\ 4 \end{bmatrix} + t \begin{bmatrix} 10 \\ 1 \\ 5 \end{bmatrix}$ does not intersect Π, and explain the geometrical significance of this result.

(4)

(Total 8 marks)

Q10.

The points A and B lie on the line l_1 and have coordinates (2, 5, 1) and (4, 1, –2) respectively.

(a) (i) Find the vector $\overrightarrow{AB}$.

(2)

(ii) Find a vector equation of the line l_1, with parameter λ.

(1)

(b) The line l_2 has equation $\mathbf{r} = \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$.

(i) Show that the point $P(-2, -3, 5)$ lies on l_2.

(2)

(ii) The point Q lies on l_1 and is such that PQ is perpendicular to l_2. Find the coordinates of Q.

(6)

(Total 11 marks)

Q11.

The lines l_1 and l_2 have equations $\mathbf{r} = \begin{bmatrix} 8 \\ 6 \\ -9 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ -3 \\ -1 \end{bmatrix}$ and $\mathbf{r} = \begin{bmatrix} -4 \\ 0 \\ 11 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ respectively.

(a) Show that l_1 and l_2 are perpendicular.

(2)

(b) Show that l_1 and l_2 intersect and find the coordinates of the point of intersection, P.

(5)

(c) The point A (– 4, 0, 11) lies on l_2. The point B on l_1 is such that $AP = BP$. Find the length of AB.

(4)

(Total 11 marks)

Q12.

The line l has equation $\mathbf{r} = \begin{bmatrix} 3 \\ 26 \\ -15 \end{bmatrix} + \lambda \begin{bmatrix} 8 \\ -4 \\ 1 \end{bmatrix}$.

(a) Show that the point P (– 29, 42, –19) lies on l.

(1)

(b) Find the acute angle between l and the z-axis.

(1)

(c) The plane Π has cartesian equation $3x - 4y + 5z = 100$.

(i) Write down a normal vector to Π.

(1)

(ii) Find the acute angle between l and this normal vector.

(4)

(d) Find the position vector of the point Q where l meets Π.

(4)

(e) Determine the shortest distance from P to Π.

(3)

(Total 14 marks)

Q13.

(a) The plane Π has equation $\mathbf{r} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$.

(i) Find a vector which is perpendicular to both $\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$.

(2)

(ii) Hence find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = d$.

(2)

(b) The line L has equation $\left(\mathbf{r} - \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}\right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \mathbf{0}$.

Verify that $\mathbf{r} = \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} + t\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ is also an equation for L.

(2)

(c) Determine the position vector of the point of intersection of Π and L.

(4)
(Total 10 marks)

Q14.

The points A and B have coordinates (2, 4, 1) and (3, 2, –1) respectively. The point C is such that $\overrightarrow{OC} = 2\overrightarrow{OB}$, where O is the origin.

(a) Find the vectors:

(i) $\overrightarrow{OC}$;

(1)

(ii) $\overrightarrow{AB}$.

(2)

(b) (i) Show that the distance between the points A and C is 5.

(2)

(ii) Find the size of angle BAC, giving your answer to the nearest degree.

(4)

(c) The point P(α, β, γ) is such that BP is perpendicular to AC.

Show that $4\alpha - 3\gamma = 15$.

(3)
(Total 12 marks)

Q15.

The points A and B have coordinates (1, 4, 2) and (2, –1, 3) respectively.

The line l has equation **r** = $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

(a) Show that the distance between the points A and B is $3\sqrt{3}$.

(2)

(b) The line AB makes an acute angle θ with l. Show that cos $\theta = \frac{7}{9}$.

(3)

(c) The point P on the line l is where $\lambda = p$.

(i) Show that

$$\dot{AP}.\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 7 + 3p$$

(4)

(ii) Hence find the coordinates of the foot of the perpendicular from the point A to the line l.

(3)

(Total 12 marks)